Mereck McGowan

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Professional Summary

Software Engineer & Machine Learning Researcher with hands-on experience building production web applications and data-driven systems using React, Python, AWS, Docker, and SQL. Founder of a software consultancy delivering full-stack solutions to real clients. Experienced in explainable machine learning, AI, automation, and applied data analysis across academic research and enterprise environments. Seeking **software engineering or machine learning roles** focused on shipping impactful systems at scale.

Education

Berry College | Mt. Berry, GA **05/2026**

Bachelor of Science, Computer Science | Business Minor

Data Structures & Algorithms, Designing Algorithms, Machine Learning, Artificial Intelligence, Building Programming Languages, Computing Systems, Web Development, Data Analysis

Experience

Berry College | Rome, GA **05 2025 – Current**

Machine Learning Researcher

- Developed fully explainable neural network models using NN-kNN, achieving comparable or improved accuracy over traditional black-box models
- Research paper submitted to **IJCAI 2026**

Berry College | Rome, GA **08 2025 – Current**

Teaching Assistant

- Supported instruction for Object-Oriented Programming and Program Design courses; assisted students with Java and abstraction concepts, graded assignments, and developed instructional materials.

Berry College | Rome, GA **09 2025 - Current**

Math and Computer Science Ambassador

- Created and hosted programs, advise students, and promote the Math and Computer Science programs.

Mohawk Industries | Calhoun, GA **05 2025 – 08 2025**

Data Analyst / Automation Developer

- Designed and implemented an automated pay audit system using **Python and SQL**, reducing manual review time and improving accuracy
- Built an asset utilization analysis system to evaluate logistics truck performance and efficiency

Freelance Developer | Asheville, NC **05 2024 - Current**

Owner

- Built and deployed full-stack React applications for clients, using Dockerized AWS infrastructure with Nginx and SSL for production-grade reliability.
- Maintained production servers and client websites, handling updates, uptime, and performance monitoring

Berry College | Rome, GA **05 2024 - Current**

Employer Engagement Specialist

- Managed and analyzed job and applicant data using **Python, Excel, and R**
- Built analytical models from Handshake and JobX data to support hiring and reporting insights
- Collaborated cross-functionally with employer engagement and career development teams

Projects

Evala | *Programming Language for Automated Testing*
github.com/mereck600/Evala

- Designed and implemented a Java-based programming language to automatically grade code and generate robust test cases

- Built a custom parser, interpreter, and test generation framework inspired by Java-like syntax

MereckOS | *Educational Operating System*
github.com/mereck600/MereckOS

- Built a simple operating system supporting a FAT12 file system
- Implemented low-level system components using C and x86 Assembly

Explainable Neural Network (NN-kNN + RL)

- Designed fully explainable neural network models combining NN-kNN and reinforcement learning
- Achieved accuracy comparable to or exceeding traditional neural networks while maintaining interpretability
- Implemented in Python with PyTorch and custom training pipelines
- Paper submitted to the IJCAI 2026 conference

Technical Skills

Languages: Python, Java, JavaScript, C++, C, SQL, R, Racket, Assembly
Frontend: React, HTML, CSS
Backend & Data: REST APIs, SQL databases, NoSQL databases, data pipelines
Machine Learning: PyTorch, neural networks, explainable ML, reinforcement learning
Cloud & DevOps: AWS EC2, Docker, Nginx, Linux, Git
Tools: GitHub, CI/CD, Excel

Involvement

Berry College Computer Science Club | *President* 2024- 2026

Berry College Men's Lacrosse | *Player* 2022 - 2024